Venkata Lakshmi Parimala Pasupuleti

Applied AI Engineer · Python / FastAPI / PostgreSQL · LLM Integration & Prompt Evaluation · Production SaaS AI

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SUMMARY

Applied AI Engineer with 9 years of production software ownership. Primary stack: Python, FastAPI, PostgreSQL, vector databases, LLM APIs -- built AgentProbe (an open-source prompt evaluation and LLM testing platform with CI regression gate, LLM-as-judge scoring, and pass@k harness) and LeaseI (an 8-agent LangGraph production AI system with FastAPI backend and RAG over structured data). Seven years at Entrata shipping B2B SaaS production modules -- Python, PostgreSQL, REST APIs -- serving 100K+ monthly active users. Daily Claude Code user. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & LLM Engineering: LLM APIs (Anthropic Claude, OpenAI, Groq, Ollama), Generative AI, LangGraph, LangChain, Agentic AI, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Prompt Engineering, Prompt Evaluation, LLM Testing, LLM-as-Judge, Hallucination Detection, Output Validation, HITL, Guardrails, AI-Assisted Development (Claude Code), Tool Calling, Structured Outputs, MCP

Backend & APIs: Python, FastAPI, REST APIs, PostgreSQL, SQL, PL/SQL, Flask, Node.js, RabbitMQ, Redis, Microservices, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), HIPAA Compliance, PHI Data Handling, Git, Linux, Agile/Scrum

Languages & Frameworks: Python, JavaScript, TypeScript, Java, SQL, React.js, Angular, Bash, HTML/CSS

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic system with FastAPI backend, RAG retrieval, prompt-engineered guardrails, and conditional routing -- designed and refined prompts to optimize AI response quality, validated generative AI outputs, and iterated on feature performance before client delivery.

Entrata (B2B PropTech SaaS · 100K+ MAU · Python / PostgreSQL / REST APIs)

Sep 2018 - Aug 2024

Senior Software Engineer

- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints in a B2B SaaS platform serving 100K+ monthly active users; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction).
- Owned and optimized 200+ PostgreSQL queries (30% performance gain) across multi-tenant schemas for financial and compliance workflows; built Python/Bash automation pipelines eliminating 80% of manual operational workload.
- Led AWS S3 document migration -- 100% PHI compliance with Libsodium-encrypted pipelines, zero security incidents; owned HIPAA-adjacent data workflows throughout tenure.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools reducing investigation time by 40%.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data integrity checks; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: Prompt Evaluation & LLM Testing Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source prompt evaluation and LLM testing platform: 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), CI-integrated regression gate blocking quality degradation on every commit, LLM-as-judge scoring with agent tracing, multi-trial pass@k harness. 15-endpoint FastAPI REST backend; multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama).
- Evaluated LeaseI across 1,240 production-quality test cases -- proving that generative AI features can be tested, measured, and iterated on systematically before and after each change.

LeaseI: 8-Agent LangGraph Production AI System

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production AI application with FastAPI backend: 8-agent LangGraph system with RAG over structured data, prompt-engineered output validation, vector database retrieval (ChromaDB), HITL guardrails on regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: HIPAA-Compliant Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, NLP-to-FHIR API translation, LLM + BERT integration; reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017